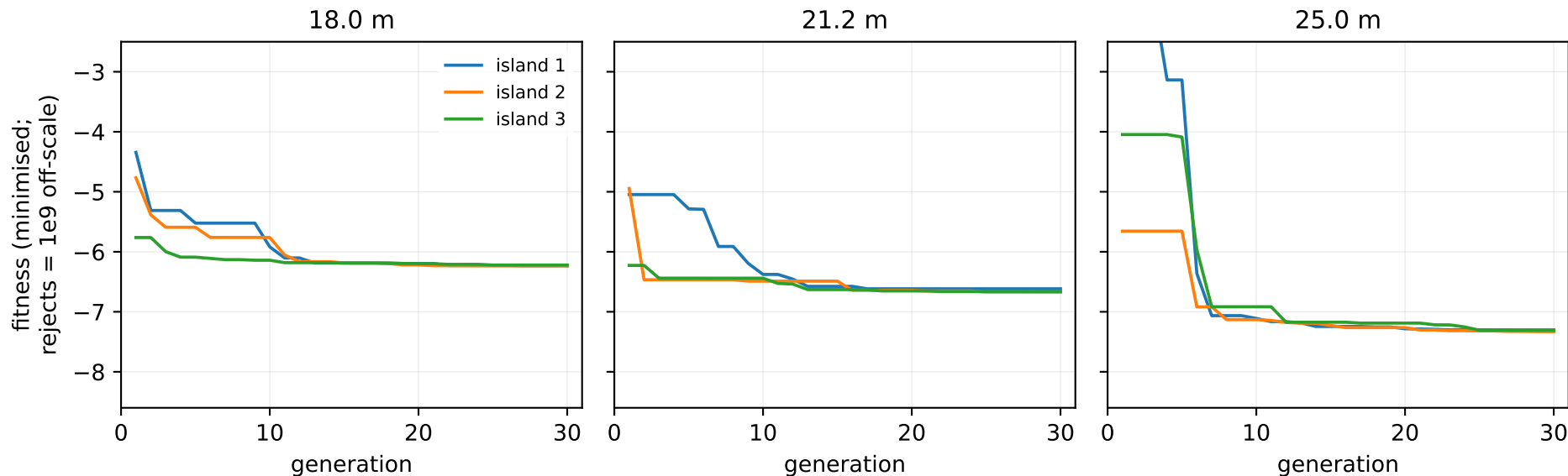


## 5 kW DE campaigns — best fitness per generation (minimised)



Three 5 kW design campaigns, one per chain length: each line is one of the three parallel search populations ('islands') — its best design per generation. Lower fitness = better design; all searches converge to a plateau — 18 m at  $-6.22$ , 21.2 m at  $-6.66$ , 25 m at  $-7.31$  — the longer chain reaches the deeper minimum. The plateaus are the edge of the design space, not a stalled search: fitness stopped improving because the remaining designs fail clearance or twist, not because the search gave up.